MINGZE JIANG

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EDUCATION

Tsinghua University *Aug 2023–Jun 2027(expected)*

B.E. in Computer Science and Technology

- **GPA: 3.9** / 4.0 **Rank: 8** / 194 **(top 5%)**
- **Standardized Tests:** GRE **339** (Quantitative 170 / Verbal 169) + 3.5 (AW)
  TOEFL **113** (Listening 30) IELTS **8.0** (Listening 9.0 / Reading 9.0)
- **Relevant Coursework:** Introduction to Artificial Intelligence (**A+**), Introduction to Complex Analysis (**A+**), Software Engineering (**A+**), Data Structures & Algorithms (**A**), Digital Image Processing (**A**), Fundamentals of Computer Graphics (**A**), Physics for Scientists & Engineers (**A**), Calculus (**A**), Discrete Mathematics (**A**), Probability and Statistics (**A**)

PUBLICATIONS & MANUSCRIPTS

- **Mingze Jiang**, Xueyang Li, John Kheir, Alec Girten, Yiyu Shi. "LightRefine-PCXR: A Lightweight Refinement Framework for Efficient Medical Device Suppression in Pediatric Chest X-Rays". *Medical Imaging with Deep Learning (MIDL)*, 2026. [*pdf*]
- Xueyang Li, **Mingze Jiang**, Gelei Xu, Jun Xia, Ching-Hao Chiu, Mengzhao Jia, Danny Chen, Yiyu Shi. "CRC-Router: Risk-Constrained Routing for Medical Agentic AI Systems". *IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*, 2026. [*pdf*]
- Kunzan Liu, Kirti Nath, Guhan Qian, Jiashu Han, **Mingze Jiang**, Li-Yu Yu, Honghao Cao, Ellen L. Kan, Linda G. Griffith, Harikesh Wong, Sixian You. "Decoding Cellular Phenotypes in Label-Free Nonlinear Imaging of Living Tissues". *SPIE Photonics West*, 2027 (Oral Presentation).
- **Mingze Jiang**, Chunqiao Li, Wenjian Yu. "A Structure-Aware Parallel Multigrid Solver for Power Grid Analysis of Flat Panel Displays". *Design, Automation and Test in Europe Conference (DATE)*, 2027 (Under review).
- Xueyang Li, Alec Girten, Peng Zeng, **Mingze Jiang**, Gelei Xu, Danny Z. Chen, Yiyu Shi, John N. Kheir. "PCXR-Copilot: Device-Aware Pediatric Chest Radiographic Analysis with Segmentation, Suppression, and Edema-Related Cues". *npj Digital Medicine* (Under review).
- Jack Tomkiewicz, Federico Presutti, Tade Marozsak, **Mingze Jiang**, Jia Zhao, Hatoon Baazim, Hernandez Moura-Silva, Eliezer Calo, Sixian You. "Single-pulse, Label-free, Volumetric Lifetime Microscopy with Near Photon-counting Photon Economy". *eLight* (Under review).

RESEARCH EXPERIENCE

Computational Biophotonics Lab, *MIT* *Cambridge, USA*

Research Intern, Advisor: Prof. Sixian You *(on-site Jul–Aug 2026) Feb 2026–present*

- **Visual Representation Learning for Biomedical Imaging**
  - Developed self-supervised representations of cell morphology and microenvironment, achieving 85.2% mean accuracy across 14 molecular markers and supporting identification of 20+ immune cell subtypes for label-free analysis of cell dynamics and metabolism in living tissues. *(SPIE Photonics West 2027, Oral Presentation)*
  - Designing in-context segmentation methods with visual foundation models for training-free annotation transfer across biomedical images, with extensions to cross-modal correspondence.
- **Multimodal Analysis for Biological Discovery in the Brain**
  - Integrating label-free SLAM microscopy with spatial transcriptomics to link imaging-derived myelin phenotypes to molecular programs associated with aging and APOE genotype.

Sustainable Computing Lab (iSURE Program), *University of Notre Dame* *South Bend, USA*

Research Intern, Advisor: Prof. Yiyu Shi *(on-site Jul–Aug 2025) May 2025–Feb 2026*

- **AI for Pediatric Critical Care *(Collaboration with Boston Children's Hospital and Harvard Medical School)***
  - Proposed LightRefine-PCXR for device suppression in pediatric chest X-rays, improving reconstruction fidelity with less than 50% of the trainable parameters and GPU memory of full fine-tuning. *(accepted at MIDL 2026, first author)*
  - Contributed to PCXR-Copilot, a chest X-ray AI system supporting pulmonary edema assessment in pediatric critical care, integrated into TWIST platform at Boston Children's Hospital. *(submitted to npj Digital Medicine)*

- **Risk-Aware Medical Agentic AI**
  - Combined uncertainty estimation with conformal risk control for reliable medical agents, achieving the best balance of risk and coverage among evaluated baselines (3.9% risk at 91.3% coverage). *(accepted at BIBM 2026, second author)*

Knowledge Engineering Group, *Tsinghua University* *Beijing, China*
Group Member, Advisor: Prof. Yuxiao Dong *(on-site) Mar 2024–May 2025*

- **Multimodal Foundation Models for Visual Understanding and Generation**
  - Explored model architectures and training strategies for multimodal visual understanding and generation.
  - As a member of the CogAgent (a vision-language model for GUI interaction) team, built a web data collection and instruction-tuning pipeline to fine-tune CogAgent-9B for calculator-assisted reasoning.

Mori Laboratory, *Nagoya University* *Nagoya, Japan*
Visiting Student, Advisor: Prof. Kensaku Mori *(on-site) Jan 2024–Feb 2024*

- **Super-Resolution for Endoscopic Imaging**
  - Investigated subpixel registration and fusion with distance-based weighting across successive frames to improve image resolution in capsule endoscopy.

SCHOLARSHIP

2025 **Comprehensive Excellence Scholarship**, Tsinghua University (Top 5%)
2024 **Comprehensive Excellence Scholarship**, Tsinghua University (Top 5%)
2023 **Freshman Scholarship**, Tsinghua University (Top 2%)
2026 **Tsinghua "TOP OPEN" Undergraduate Overseas Academic Research Program** (Selected Awardee)

HONORS & AWARDS

2022 **National Gold** in 36th Annual AAPT (**American** Association of Physics Teachers) Physics Bowl Contest
2022 **Global Gold** in **British** Physics Olympiad Senior Physics Challenge
2022 **Global Silver** in **British** Physics Olympiad Competition (BPHO R2)
2022 **National Silver, Global Top 150** in **Canada** SIN (Sir Isaac Newton) Contest
2022 **Canadian Top 20, China National Silver** in CAP (**Canadian** Association of Physicists) High School Physics Prize
2021 **Top Gold** in **British** Physics Olympiad Competition (BPHO R1)

SERVICE & LEADERSHIP

- **Technology Equity and Digital Divide Research** *(Core Member)* *Jul–Aug 2024*
  Studied regional technology adoption; published findings and earned Tsinghua's *Student Social Practice Silver Award*.
- **Student Academic Development Association, Tsinghua** *(Core Member)* *Mar 2025–Jun 2026*
  Promoted the Buddy Program and campus campaigns; content reached 1,000+ views.
- **Inspire Epistolary Mentorship Program, Tsinghua** *(Volunteer)* *Feb–Jul 2025*
  Mentored students in under-resourced regions through handwritten letters.
- **Yinhe Correspondence Program, Tsinghua** *(Volunteer)* *Sep 2024–Jan 2025*
  Supported hearing-impaired students at Beijing Qiyin Experimental School through letters.

RESEARCH INTERESTS

AI for Biomedicine: AI for Biology, AI for Health, Computational Biology
Visual and Multimodal Learning: Computer Vision, Multimodal Learning, Representation Learning